Project: Open Port Scan and Service Mapping DEV-352

Results Report

1. Objective

The objective of this project was to scan **15 authorized IP addresses** and identify:

- Host availability
- Open TCP ports
- Services running on open ports
- Service/version information where detected

The scanning process was performed using **Nmap** and automated through a **Bash script**.

2. Tools and Technologies

Tool / Technology	**Purpose**
Nmap 7.98	Port scanning and service detection
Bash Script	Automated scanning of multiple IP addresses
Linux/Ubuntu	Scanning environment
TCP Connect Scan	Detection of reachable TCP ports
Nmap `-sV`	Service and version identification
Nmap `-Pn`	Scanning hosts without relying on ICMP discovery

3. Scan Results

The 15 authorized IP addresses were scanned to identify reachable hosts, open TCP ports, and running services.

#	IP Address	Open Ports	Services Detected	Status
1	64.23.130.208	22, 54321	SSH, MariaDB	**Completed**
2	157.230.47.60	22, 53, 4646, 8301, 8500, 8502, 8503, 8600, 9000, 9001	SSH, DNS, HTTP, Consul/MinIO-related services	**Completed**
3	159.223.62.168	22	SSH	**Completed**
4	139.59.245.244	22	SSH	**Completed**
5	143.198.94.161	22	SSH	**Completed**
6	188.166.250.175	22, 53, 80, 443, 7212, 7213, 9111, 9112, 9180	SSH, DNS, HTTP/HTTPS, PostgreSQL, Redis, HTTP services	**Completed**
7	139.59.117.80	22, 53, 80, 443, 7212, 7213, 9111, 9112, 9180	SSH, DNS, HTTP/HTTPS, PostgreSQL, Redis, HTTP services	**Completed**
8	134.209.107.38	22	SSH	**Completed**
9	146.190.97.129	22	SSH	**Completed**
10	128.199.134.178	22	SSH	**Completed**
11	167.172.66.204	22	SSH	**Completed**
12	139.59.113.219	22, 80, 443, 8443	SSH, HTTP/HTTPS	**Completed**
13	159.89.196.66	22	SSH	**Completed**

14	139.59.230.32	22	SSH	**Completed**
15	139.59.99.241	22, 443	SSH, HTTPS	**Completed**

4. Overall Port Summary

Port / Service	**IPs Exposing It**	**Count**
22 – SSH	All 15 IPs	**15**
443 – HTTPS	188.166.250.175, 139.59.117.80, 139.59.113.219, 139.59.99.241	**4**
80 – HTTP	188.166.250.175, 139.59.117.80, 139.59.113.219	**3**
53 – DNS/TCP	157.230.47.60, 188.166.250.175, 139.59.117.80	**3**
7212 – PostgreSQL	188.166.250.175, 139.59.117.80	**2**
9111 – Redis	188.166.250.175, 139.59.117.80	**2**
54321 – MariaDB	64.23.130.208	**1**
8443 – Alternative Web Service	139.59.113.219	**1**
Other Application Ports	Mainly **157.230.47.60, 188.166.250.175, 139.59.117.80**	**Multiple**

Port Exposure Overview

- **15/15** IPs expose SSH on port **22**.
- **4** IPs expose HTTPS on port **443**.
- **3** IPs expose HTTP on port **80**.
- **3** IPs expose DNS/TCP on port **53**.
- **2** IPs expose PostgreSQL on port **7212**.

- **2** IPs expose Redis on port **9111**.
- **1** IP exposes MariaDB on port **54321**.

5. Exposure Assessment

Based on the number and type of publicly reachable services, the targets were grouped by exposure level.

Exposure Level	IP Addresses	Count
High Exposure	157.230.47.60, 188.166.250.175, 139.59.117.80	**3**
Moderate Exposure	139.59.113.219, 139.59.99.241	**2**
Review Required	64.23.130.208	**1**
Limited Exposure	Remaining 9 IPs	**9**

Note: This classification represents observed network service exposure. It does **not** confirm that a host or service is vulnerable.

6. Important Findings

1. **SSH (port 22)** was the most commonly exposed service and was found on all 15 targets.
2. **157.230.47.60** had the largest number of open ports, including Consul/HTTP-related and other application services.
3. **188.166.250.175** and **139.59.117.80** exposed multiple services, including web services, PostgreSQL, Redis, and other application services.
4. **64.23.130.208** exposed MariaDB on port **54321**, which should be reviewed to confirm whether external access is required.
5. **139.59.113.219** exposed SSH, HTTP, HTTPS, and port **8443**.
6. Several application ports could not be confidently identified and should be reviewed to determine their purpose.

7. Security Observation

The scan provides a **network service inventory and attack-surface overview**.

The following systems had the highest exposed service surface and should receive higher review priority:

1. **157.230.47.60**
2. **188.166.250.175**
3. **139.59.117.80**

Database, Redis, management, and application services exposed directly to the network should be reviewed to ensure that only required services are publicly accessible.

An **open port does not by itself indicate a vulnerability**. Further security testing would be required to determine whether an exposed service is vulnerable or incorrectly configured.

8. Conclusion

The project successfully identified the **open TCP ports and associated services for all 15 authorized IP addresses**.

The results show that **SSH is the most commonly exposed service**, while a smaller number of systems expose web, database, Redis, storage, and infrastructure-related services.

The generated results provide a useful **network service inventory and attack-surface overview** that can support further security assessment and firewall review.

Overall Result: Scan Completed Successfully.